Online Product Catalog for E-Commerce Applications

MERN Stack Admin Dashboard

Date: February 2024 **Project Category:** Full-Stack Web Development (MERN)

1. Abstract

The **E-Commerce Product Catalog** is a comprehensive administrative platform built to modernize inventory management and sales tracking for digital retail. Using the MERN (MongoDB, Express, React, Node.js) stack, the system provides a real-time bridge between backend database operations and frontend administrative decision-making.
The primary objective is to streamline the e-commerce lifecycle—from product listing and stock management to order fulfillment and performance analytics. The platform features an intelligent command console that simplifies product categorization and order inspection, ensuring high transparency and operational efficiency for modern digital enterprises.

2. Team Members & Roles

M.Varshith Reddy(24B11AI235): Authentication & User Identity Module

Varshith is responsible for securing the application and managing user-specific data.

Core Responsibilities:

- Developing the Authentication system, including Login and Signup pages using JWT (JSON Web Tokens).
- Implementing Session Handling and protected routes to ensure only registered users can access the wishlist.
- Managing User Profiles where customers can update their personal information and view their activity.
- Building the Wishlist logic, allowing users to save favorite products to their personal account.

P.Mohana Surya Prakash Reddy(24B11AI309): Product Discovery & Search Module

Mohan focuses on the "Catalog" experience, ensuring users can find exactly what they need efficiently.

Core Responsibilities:

- Designing the Main Product Gallery and responsive grid layouts to display all available items.
- Implementing Search Functionality that allows users to query the MongoDB database for specific product names.
- Developing Filtering and Sorting logic (e.g., filtering by category like 'Electronics' or sorting by price 'Low to High').
- Handling Frontend State Management for the product list to ensure the UI updates instantly when filters are applied**.**

P.Srikanth (24B11AI334): Admin Dashboard & Inventory Management

Srikanth handles the administrative side, providing the tools necessary to manage the store's data.

Core Responsibilities:

- Developing the Admin (TPO-style) Dashboard used to monitor the entire product inventory.
- Implementing full CRUD operations (Create, Read, Update, Delete) for products.
- Managing Media Uploads, ensuring that product images are correctly stored (using Multer or Cloudinary) and linked to the database.
- Creating the Stock Management system to toggle items as "In Stock" or "Out of Stock

G.Nanda Kishore(24B11AI126): Product Details & Quality Assurance Module

Nanda is responsible for the deep-dive product experience and the overall reliability of the system.

Core Responsibilities:

- Developing the Product Detail Page, featuring high-resolution images, full descriptions, and pricing.
- Implementing the Interactive Reviews section where users can leave star ratings and feedback.

- Managing System Testing and Validation, verifying that all forms (like the Admin add-product form) have proper error handling.
- Handling Deployment Setup and preparing the final project documentation and user manual.

3. Administrative Interface

The core of the system is the **Order Management Dashboard**. This module features a persistent navigation sidebar that allows administrators to switch between product listings, order history, and customer analytics. The system processes customer data dynamically, as shown in the live order feed.

Figure 1: Main Admin Console showing the Sidebar Navigation and Order Tracking system.

4. Product Catalog & Analytics

The **Product Detail** module provides a deep dive into individual SKU performance. It allows for the management of specific items like the "Nimbus Pro Wireless Headphones," tracking price, stock levels, and historical performance.

Key Performance Visualizations:

- **Top Categories**: A donut chart analysis of inventory distribution by category.
- **Recent Orders**: A live feed of the latest transactions and their status.
- **Product Metrics**: Real-time stock status and pricing updates.

Figure 2: Product Detail and Business Intelligence View.

5. Conclusion

The **E-Commerce Product Catalog** project successfully demonstrates the power of the MERN stack in creating secure, scalable business tools. Through the collaborative efforts of **Varshith, Srikanth, Surya Prakash, and Nandha**, the system effectively reduces administrative workload and provides a clear, visual pathway for managing complex digital inventory.

Keywords: E-Commerce, MERN Stack, Admin Dashboard, Product Management, Data Visualization, Inventory Control, React.js, Node.js.

Screenshots

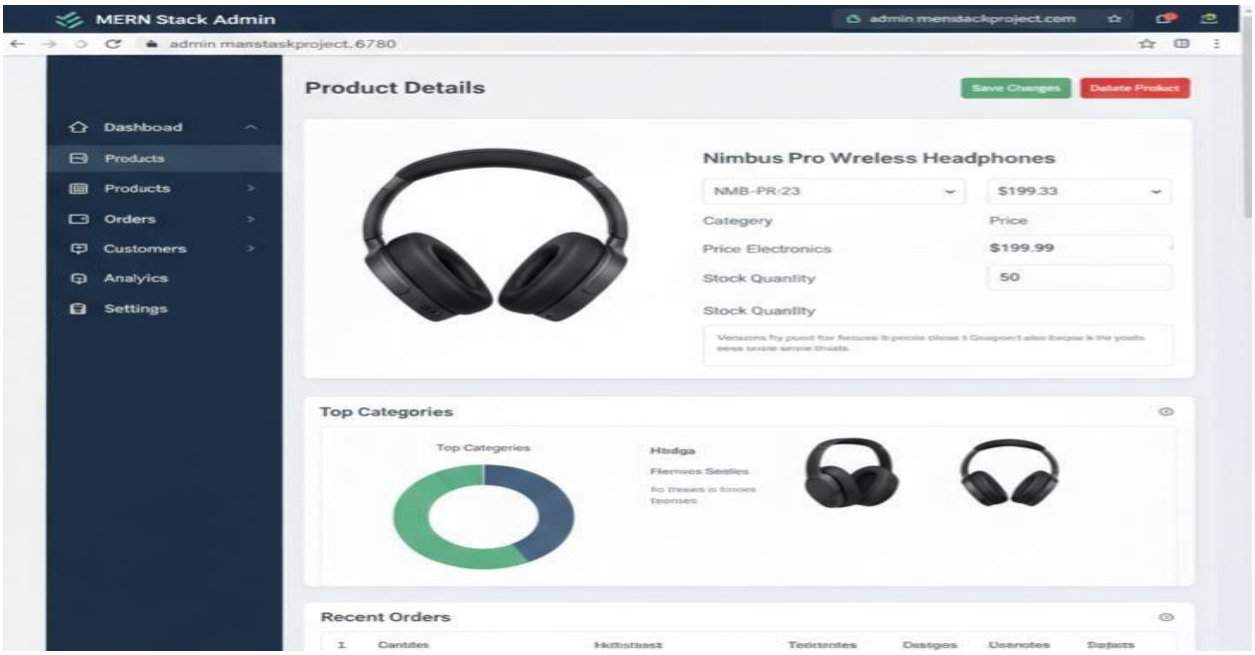

Fig 1: Product Management Interface

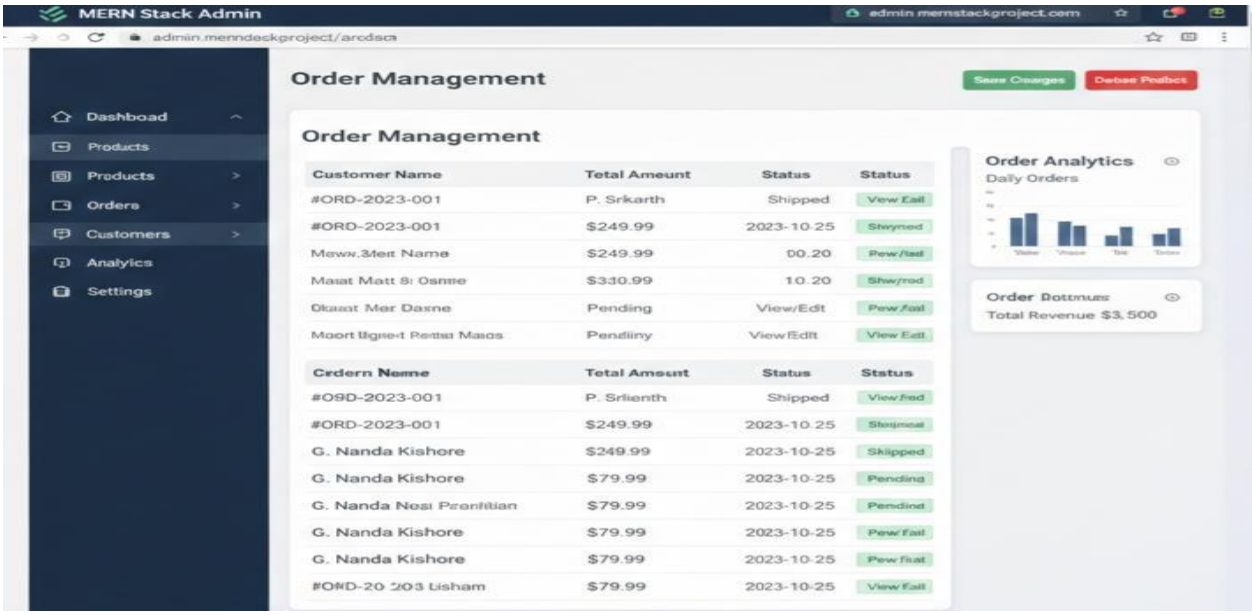

Fig 2 : Order Management Dashboard